Andrea Muolo

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github.com/amuolo

EXPERIENCE

Postdoctoral Research Fellow
Hebrew University of Jerusalem

Oct 2019 - present Jerusalem, Israel

- Exact factorization techniques in molecular physics
- Genuine predictions on the molecular Berry phase
- Fellow of the Swiss National Science Foundation (SNSF)

Research and Teaching Assistant
ETH Zürich

Dec 2018 - Sept 2019 Zürich, Switzerland

- Formulation of many-body Dirac Equation and relativistic quantum mechanics.
- Student supervision: research project on "Development of pre-Born–Oppenheimer Tensor Network Algorithms"
- Teaching activities: Thermodynamics and Advanced Quantum Chemistry

EDUCATION

Doctor of Sciences
ETH Zürich

Nov 2014 – Dec 2018 Zürich, Switzerland

Explicitly Correlated Gaussians and the Quantum Few-Body Problem
Advisor: Prof. Dr. M. Reiher
Examiner: Prof. Dr. J. Richardson

Second level Master's Degree in Science and Technologies
Scuola Universitaria Superiore di Pavia (IUSS)

Oct 2009 – Sept 2014 Pavia, Italy

Scientific specialisation characterised by a high level ongoing education
I passed exams in: History of Science, Bioethics, Law, Symmetries, and fundamental constants: the frontiers of Physics, Laser and Photonics, Steps to make a space mission, Nanotechnologies

Master's Degree in Physical Chemistry
University of Pavia and University of Cambridge

Oct 2012 – Sept 2014 Pavia, Italy – Cambridge UK

A classical dynamic that conserves the quantum Boltzmann distribution
Supervisor: Prof. Dr. S. Althorpe
Final grade: 110/110 *cum laude*

Bachelor's Degree in Chemistry
University of Pavia

Sept 2009 – Sept 2012 Pavia, Italy

Molecular Memories: rotaxane- and catenane-based molecular devices
Supervisor: Prof. Dr. L. Fabbrizzi
Final grade: 110/110 *cum laude*

MOST PROUD OF

Swiss National Science Foundation (SNSF) Fellowship
Research Grant, October 2019 - present

Best Poster Presentation
MOLIM General Meeting, April 2018, Budapest.

PUBLICATIONS SELECTION OF

Phys. Rev. A, 102, 022803 (2020)
Analytically projected, rotationally symmetric, explicitly correlated Gaussians with one-axis shifted centers

J. Chem. Phys., 152, 204103 (2020)
Nuclear-electronic all-particle density matrix renormalization group

J. Chem. Phys., 151, 154110 (2019)
H_3^+ as a five-body problem described with explicitly correlated Gaussian basis sets

J. Chem. Phys., 149, 184105 (2018)
Explicitly correlated Gaussian functions with shifted-center and projection techniques in pre-Born-Oppenheimer calculations

Chimia, 70, 244 (2016)
New Approaches for *ab initio* Calculations of Molecules with Strong Electron Correlation

J. Chem. Phys., 142, 134103 (2015)
Boltzmann-conserving classical dynamics in quantum time-correlation functions: "Matsubara dynamics"

COMPUTER SKILLS

- C++ Libraries: Armadillo, Boost, Eigen
- C++ 17
- C# 8
- Git
- Linux
- Latex
- Mathematica
- Matlab
- MPI 3.1
- OpenMP 4.0
- Python

COMPETENCES

Knowledge transfer | Teamwork | Learning new things | Problem solving | Natural sciences | Numerical methods | Detail oriented | Critical thinking | Multitasking | Public speaking/presentation | Empathic listening | Flexibility skills | Student supervision

LANGUAGES

- Italian (mother tongue)
- English (C1)
- German (A2)
- French (A2)
- Spanish (A2)

INTERESTS

- Sailing, rowing, swimming, dancing
- Diving into different points of view
- Cooking for dinner with friends
- Studying financial risk management